Course Code: CSExxxxx	**Course Name:** Advanced Data Science	
Credits: 3 (2-0-2)	**Contact Hours:** 2 hours theory & 2 hours lab per week	
Batch: 2024, 5th Sem **Academic Year:** 2026-27	**Semester Duration:** 27th Jul 2026 to 24th Nov 2026	
Course Faculty: 1. Dr Hirdesh Kumar Pharasi [*hirdesh.pharasi@bmu.edu.in, 97755 50708 and 89, IV Floor, E2 Building*] 2. Prof. Anantha Rao [anantha.rao@bmu.edu.in*, 98681 60550 and 35, IV Floor, E2 Building*]	**Course Coordinator:** **Name:** Dr Hirdesh Kumar Pharasi **Email:** *hirdesh.pharasi@bmu.edu.in* **Office:** 89, IV Floor, E2 Building	

Aim of the course: The Advanced Data Science course aims to equip students with the knowledge and practical skills required to design, develop, deploy, and manage industry-scale data science solutions. The course moves beyond predictive modelling to focus on decision-centric analytics, experimentation, causal reasoning, scalable machine learning systems, MLOps, Responsible AI, and modern foundation models. Through hands-on projects and real-world case studies, students will learn to build reliable, interpretable, and production-ready AI systems capable of supporting data-driven decision-making across diverse application domains.

Course Overview and Context: Advanced Data Science is an elective course offered to third-year B.Tech students specialising in Data Science and Artificial Intelligence. The course focuses on the complete data science lifecycle, covering problem formulation, data engineering, scalable machine learning, experimentation, deployment, monitoring, and Responsible AI. Students will gain hands-on experience with modern tools and frameworks such as Apache Spark, MLflow, Docker, Kubernetes, and transformer-based models while learning MLOps, causal inference, Explainable AI, and recommendation systems. A semester-long industry-oriented capstone project enables students to build, deploy, and monitor end-to-end AI solutions using industry best practices, preparing them for careers in data science, AI engineering, and research.

Course Outcomes (CO): At the end of the course the students should be able to do the following:

	Course Outcome Statement	**Mapped Component**	**Associated Activities / Assessment**
CO1	Understand advanced data science concepts, including problem framing, experimentation, causal inference, scalable data engineering, MLOps, and Responsible AI for solving real-world decision-making problems.	Employability	• Theory sessions • Case studies • Research paper discussions • Quizzes
CO2	Apply advanced data science techniques and modern tools to build, optimise, deploy, and monitor scalable machine learning solutions using distributed computing, MLOps frameworks, and cloud-based platforms.	Skill Development	• Theory sessions • Case studies • Research paper discussions • Quizzes

CO3	Design, evaluate, and deploy end-to-end data science solutions by integrating experimentation, explainable AI, ethical AI principles, and production-grade machine learning workflows to address complex real-world challenges.	Entrepreneurship, Skill Development	• Semester-long industry capstone project • Model deployment • Performance evaluation • Project presentation, demonstration, and viva

Prerequisite: if any

1. Basic knowledge of Python programming
2. Understanding of fundamental Data Science Concepts

Topics of the course:

Topics	Number of Sessions (48)	Course Outcomes
Problem Framing and Business Understanding: Decision-centric data science, business objectives, KPI and metric design, data-driven decision making	3 (2 Th + 1 Lab)	CO1
Experimentation and Causal Inference: Hypothesis testing, A/B testing, uplift modelling, causal inference in economic applications, propensity score matching, regression discontinuity, difference-in-differences	6 (3 Th + 3 Lab)	CO1, CO2
Data Engineering for Advanced Analytics: ETL pipelines, workflow orchestration, Apache Spark, Spark SQL, distributed computing, stream processing, data warehousing	6 (2 Th + 4 Lab)	CO1, CO2
Advanced Machine Learning: Advanced feature engineering, feature selection, dimensionality reduction, ensemble learning, gradient boosting, hyperparameter optimisation, model calibration	8 (3 Th + 5 Lab)	CO1, CO2
Decision Intelligence: Recommendation systems, reinforcement learning fundamentals, optimisation techniques	5 (2 Th + 3 Lab)	CO1, CO2
MLOps and Production AI Systems: CI/CD for ML, model deployment, Docker, Kubernetes basics, MLflow, Kubeflow, API development, model monitoring, data drift, concept drift, automated retraining pipelines	8 (3 Th + 5 Lab)	CO2, CO3
Responsible and Explainable AI: Fairness in AI, bias detection and mitigation, privacy-preserving machine learning, AI governance, Explainable AI using SHAP and LIME	5 (2 Th + 3 Lab)	CO2, CO3
Modern AI Models: Transfer learning, attention mechanisms, transformers, representation learning, fine-tuning pretrained models	5 (2 Th + 3 Lab)	CO2, CO3
Industry Capstone Project: End-to-end data science solution involving experimentation, modelling, deployment, monitoring, explainability, and Responsible AI	2 (2 Lab)	CO3

CO/PO Mapping:

CO/PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	3	1	2	1			2				1	3	2		
CO2	2	2	3	2	3			1			2	1	3	3		
CO3	1	2	3	3	2	2	1	3	2	2	3	2	3	3		

Learning Resources:

Textbooks

1. Taddy, M., Hendrix, L., & Harding, M. C. (2023). Modern Business Analytics: Practical Data Science for Decision Making. McGraw-Hill Education.
2. Jain, P., & Sharma, P. (2018). Behind Every Good Decision. Harper Business.
3. Beaulieu, A. (2020). Learning SQL: Generate, Manipulate, and Retrieve Data (3rd ed.). O'Reilly Media.
4. Géron, A. (2023). Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems (3rd ed.). O'Reilly Media.
5. Dobber, D. C., & Gross, C. (2020). Python Data Visualization Workshop: A Practical Guide to Creating Effective Data Visualizations with Python. Packt Publishing.
6. Lakshmanan, V., Robinson, S., & Munn, M. (2020). Machine Learning Design Patterns: Solutions to Common Challenges in Data Preparation, Model Building, and MLOps. O'Reilly Media.
7. Molnar, C. (2024). Interpretable Machine Learning: A Guide for Making Black Box Models Explainable (2nd ed.).

Reference Books

1. VanderPlas, Jake. *Python Data Science Handbook: Essential Tools for Working with Data*. 2nd ed. O'Reilly Media, 2022.
2. Nathan George. *Mastering Design of Experiments with Python*. Packt Publishing, 2024.
3. Jules S. Damji, Brooke Wenig, Tathagata Das, and Denny Lee. *Learning Spark: Lightning-Fast Data Analytics*. 2nd Edition, O'Reilly Media, 2020.
4. Chip Huyen. *Designing Machine Learning Systems*. O'Reilly Media, 2022.
5. Emmanuel Ameisen. *Building Machine Learning Powered Applications*. O'Reilly Media, 2020.

Video lectures/ MOOC:

1. https://www.youtube.com/watch?v=_ZEWDGpM-vM&list=PLYCpMb24GpOC704uO9svUrihl-HY1tTJJ

Online Resource:

1. https://github.com/jakevdp/PythonDataScienceHandbook/tree/master/notebooks
2. https://github.com/TaddyLab/MBA

Experiential Learning Component:

The capstone project is the primary experiential learning component of this course. Students will solve real-world data science problems by applying the complete data science lifecycle, including business understanding, experimentation, modelling, deployment, monitoring, and Responsible AI practices. They will use industry-standard tools such as Apache Spark, MLflow, Docker, Kubernetes, and cloud platforms to build scalable and production-ready solutions. Laboratory assignments will reinforce concepts in data engineering, MLOps, Explainable AI, and modern machine learning. Students are expected to spend approximately 15–20 hours on programming assignments and 20–25 hours on the capstone project. The capstone project carries 40% weightage, while programming assignments contribute 30%, resulting in a total experiential learning weightage of 70%. This hands-on approach prepares students to develop and deploy industry-grade data science solutions.

Assessment Pattern: The final grade will be determined by the marks or grades earned during the project's phase-wise evaluations and the end-term assessment. Grading will be conducted using the relative grading method outlined in the university's academic regulations. To be eligible for grading, students must achieve a minimum of 40% of the total marks upon completing all assessments listed in the table below:

Evaluation Component	**Weightage (%)**	**Evaluation Schedule/Timeline**	**Rubrics/Remarks**	**CO mapping**
Assignment	20%	One week after the mid-term examination	Assignments will be evaluated offline during the laboratory sessions. Students will present and explain their solutions, followed by a theory viva to assess their conceptual understanding.	CO1, CO2
Mid-Semester Examination	20%	As per the academic calendar/Date-sheet	Written Examination	CO1, CO2
Quiz	20%	1st week of November	MCQ and problem-solving questions covering concepts and practical applications	CO1, CO2
End Term Project Evaluation	40%	As per academic calendar/Date-sheet	Students are required to submit a comprehensive project report along with the coding solution, followed by a presentation, demonstration, and viva examination. • Pre-submission-10% • Project-30%	CO1, CO2, CO3
– All evaluations will be based on the work presented by the students as well as the questions asked, or the problems given to code. – Cases of AI-generated code or plagiarism will be taken seriously and reported according to the university's policy on Unfair Means (UFM). It is essential that all work is original and adheres to academic integrity.				–

– There is a mandatory requirement to upload the project to a public repository on GitHub.	

Student Responsibilities:

- Attend lectures and do the work Lab Assignments as per instructions.
- Participate in the discussions/assignments held during classes.
- Check announcements on the LMS and emails regularly.
- Submit the assigned task on time.
- Regularly check marks on the LMS to ensure they are up to date.
- Participate in class and take necessary actions to grasp the material. Asking questions is encouraged.
- Communicate any concerns by speaking directly with the instructor.

Attendance Policy: Students are expected to attend classes regularly. Failure to follow the classes regularly and adhere to the expected attendance percentage will result in losing quiz/lab marks and a reduction of the grade as per the University's grading policy.

Recourse Examination Policy: In case a student fails the course, a one-time recourse is permitted as per the academic regulations of the University. Recourse is allowed only for the End Semester examination.

Make-up policy: No make-up exam will be conducted for unexcused absences. The faculty needs to be informed in advance in case the student is not going to appear for any evaluation component, and it is at the discretion of the faculty to sanction makeup for an evaluation component.

Behavior Expectations: No mobile phones and other distractive gadgets are permitted in the class.

Academic Dishonesty/Cheating/Plagiarism: Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.